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Implementation of error detection and correction using hamming code

Aim:-

To write a program to implement error detection and correction using hamming code concept and make a test run to input data stream and verify error.

Code:-

```
def redundant_bits(m):
    r=0
    while 2**r < m+r+1:
        r+=1
    return r

def encode(data):
    m=len(data)
    r= redundant_bits(m)
    print("Number of data bits:",m)
    n = m+r
    h= ['0'] * (n+1)
    j=0
    for i in range(r):
        p= 2**i;
        parity=0
```

```
print("Redundant bit positions: ", [2**i for i in
                                    range(r)])
print("Hamming code: ", ''.join([h[i:]]))
return ''.join h[1:])

def check(h):
    n = len(h)
    r = 0
    while 2**r <= n:
        r += 1
    error = 0
    print("\n Parity Checking:")
    for i in range(r):
        p = 2**i
        parity = 0
        for j in range(1, n+1):
            if j & p:
                parity ^= int(h[j-1])
        print("position", p, ":", parity)
        if parity != 0:
            error += p
        else:
            print("Error has occured")
            print("Error position:", error)
    return h
```

```
def remove_parity(h):
    data = ''
    for i in range(1, len(h)+1):
        if (i-1):
            data += h[i-1]
    return data

def binary_to_text(data):
    text = ''
    for i in range(0, len(data), 8):
        byte = data[i:i+8]
        text += chr(int(byte, 2))
    return text
print("Data: " text)

channel = encode(binary)
with open("Channel.txt", "w") as f:
    f.write(channel)

with open("Channel.txt", "r") as f:
    received = f.read()

if change == input("Do you want to
                    change")
```

```
if choice.lower() == "yes":
    pos = int(input())
    if pos >= 1 and pos <= len(received):
        h = list(received)
        if h[pos-1] == '0':
            h[pos-1] = '1'
        else:
            h[pos-1] = '0'
        received = ''.join(h)
print(data)
if output = binary_to_text(data)
print('output')
```

Result:-

Thus the program is implemented using python for error correction using hamming code.